Session-VI: Groundwater Draft and Balance: October 1st week

Program Schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	10 min.	Recap.	Small group discussion	List of fifth session contents, and visuals used in the seventh session.	Participants recollect the contents discussed in fifth session.
2	60 min.	Estimation of groundwater draft and balance by end of October.	Large group discussion	Charts, markers, visuals, and small pieces of paper.	Participants calculate the groundwater draft for June-October and the groundwater balance by the end of October.
3	45 min.	Estimation of crop-water requirement for Rabi crops.	Large group discussion, and small group discussion	Charts, markers and calculators.	Participants estimate the water requirement for different crops. Participants estimate the groundwater draft for Nov-May.
4	25 min.	Estimation of recharge through projected rainfall.	Demonstration, and large group discussion	10 tr10 transparent glasses, big tumbler, 10 years (Nov-May) rainfall data, calculator, data on secondary water bodies, charts and markers.	Participants discuss and estimate groundwater recharge through projected rainfall for Nov-May.
5	20 min.	Estimation of recharge through secondary water-bodies.	Large group discussion	Charts and markers.	List the secondary water bodies and discuss the process of estimation of groundwater recharge.
6	15 min.	HESA-Estimation of groundwater balance by end of May.	Large group discussion	Charts, markers, charts of previous session and calculators.	Participants analyze the relationship between amount of recharge and draft and estimate the projected balance by the end of May.
7	15 min.	Short Study- Water quality analysis	Demonstration	Transparent glasses, alum, drinking water.	Participants discuss the importance water analysis.
8	15 min.	Short Study: Cooperation among committee members.	Demonstration	Water.	Participants discuss the need for cooperation among committee members in a GMC/HUN.
9	15 min.	Group dynamics: Communication distortion.	Large group discussion	Charts and markers.	Participants identify the need to disseminate the correct information.
10	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, the participants will:

- Recap the fifth session content,
- Estimate groundwater draft and balance for the end of October,
- Estimate the crop water requirement for Rabi crops,
- Estimate the recharge through projected rainfall (Nov – May),
- Estimate the recharge through secondary water bodies, and
- Discuss about communication distortion.

Process

1. Recap Time: 10 min.

Objective : Participants will recap the learning of fifth session.

Content : Session five contents:

- Estimation of total rainfall received.
- Estimation of HU area.
- Estimation of groundwater recharge.
- Understand the importance of data accuracy.
- Short Study- Water level fluctuations (SWL, PWL Vs. Rainfall).
- Homework- Data collection for estimation of groundwater draft, Rabi crop plans, and secondary water bodies.

Method : Small group discussion

Materials : List of fifth session contents, and visuals used in the fifth session.

Process

- After formal welcome, start the session with FWS pledge.
- Request each farmer group to share at least one topic discussed in fifth session.
- Encourage each group to share the session content.
- Display the visuals used in fifth session at appropriate place so that farmers can see and recollect the contents discussed.

2. Estimation of groundwater draft and balance by the end of October Time: 60 min

Objective : Participants estimate the groundwater draft.

Content :

- Data required for estimation of groundwater draft.
- Estimation of groundwater draft.

Method : Large group discussion

Material : Charts, markers and small pieces of paper.

Process

- Ask participants about the home work and request them to present the data in a large group.
- Consolidate the data and share the data required for estimation of groundwater draft. The data includes:

- Average discharge of the bore well: A
- Average pumping days: B
- Average pumping hours per day: C
- Number of Bore wells: D

Total Discharge (E) =A x B x C x D Liters.

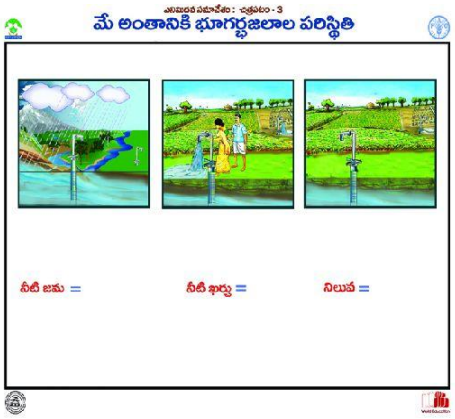




- Accordingly, estimate the groundwater draft for June to October.
- Next, remind the participants about the groundwater recharge estimated in previous session.
- Then, ask the participants to estimate the groundwater balance by the end of October.

Groundwater balance (June –October) = Groundwater recharge (June-October) – Groundwater draft (June-October).

- Use the following visual and plot the bar chart to help participants understand how much rain water recharged into the ground and how much water used for the kharif.



3. Estimation of crop water requirement for Rabi crops Time: 45 min.

Objective : Participants will estimate the crop water requirement for Rabi crops.

Content : Crop water requirement for Rabi crops.

Method : Large group discussion and small group discussion

Material : Charts, markers and calculators.

Process

- Ask participants in large group about the crops and the area proposed under different crops in their villages. (Remind the homework).
- Encourage the participants to fill the area under each crop in different habitations of the Hydrological Unit on the chart. (Keep the following format ready before conducting the session).

S.No.	Name of crop / Name of the habitation	A (acres)	B (acres)	C (acres)	D (acres)	E (acres)	F (acres)
1							
2							
3							
4	Total						

- Read aloud the total area (acres) proposed under each crop in the Hydrological Unit.
- Next, share these words with participants: "We know the types of crops and total acreage. Now, let us discuss about the water required for these planned crops."
- Encourage participants to discuss the process of estimation of water requirement for a crop.
- Take one crop as example and estimate the water requirement for one acre.
- For example, estimate the water requirement for chilly crop per acre.
    - Number of irrigations required for the crop.
    - Time taken for one irrigation (hours).
    - Discharge per hour (liters per hour).
- Let the chilly crop requires six irrigations at the rate of eight hours per irrigation and the average discharge of the borewell is 16,500 liters per hour.
- Quantity of water required for one irrigation:
    - = Discharge per hour X time taken (hours).
    - = 16500 X 8.
    - = 1,48,000 liters.
- Total quantity of water required for the crop:
    - = Water required per irrigation X No. of irrigations.
    - = 1,48,000 X 6.
    - = 8,88,000 liters = 888 cubic metres.
- Next, request participants to work in small groups to estimate water requirement for all crops.
- Based on the number of Rabi crops, encourage each group to select two to three crops to estimate water requirement.
- Request each group to present their group work to large group.
- Consolidate group presentations and share the total water requirement for entire HU.

4. Estimation of recharge through projected rainfall Time: 25 min.

Objective : Participants will estimate the recharge through projected rainfall.

Content :

- Recharge through projected rainfall.

Method : Demonstration and large group discussion.

Material : 10 transparent glasses, big tumbler, 10 years (Nov-May) rainfall data, calculator, data on secondary water bodies, charts and markers.

Process

- Remind the participants once again about the groundwater balance by the end of October and water requirement for Rabi crop plans.
- Explore the other ways of meeting water demand if the groundwater balance by the end of October is not adequate.
- Ask participants about the expected rainfall during Nov. to May.
- Discuss the estimation process of average projected rainfall using the following demonstration:
    - Take 10 transparent glasses.
    - Fill each glass with water according to rainfall received in each year.
    - Each glass represents one year rainfall (Nov – May).
    - Empty the water from all the glasses into one big tumbler.
    - Again fill the glasses equally with the water.
    - Now the water in each glass represents the average rainfall which would occur for the coming Rabi season.

5. Estimation of recharge through secondary water bodies Time: 20 min.

Objective : Participants will estimate the recharge through secondary water bodies.

Content : Session content:

- Recharge through secondary water bodies.

Method : Large group discussion

Material : data on secondary water bodies, calculator, charts and markers.

Process

- Encourage the participants to share the data on secondary water bodies.
- Ask the participants how to estimate the recharge from secondary water bodies.
- Share the process of estimation of recharge from secondary water bodies.

Tips to facilitator

Recharge from secondary water bodies:

= 1.4/1000 X Area of the water body (acres) X 4000 X 0.6 X No. of day's water stored.

6. HESA-Estimation of groundwater balance by the end of May Time: 15 min.

Objective : Participants estimate the projected groundwater balance (Nov-May).

Content : projected groundwater balance (Nov-May).

Method : Large group discussion.

Materials : Charts, markers, charts of previous session and calculators.

Process

- Ask the participants in large group about the amount of groundwater available with us.
- Encourage any one of the participants to explain the process of estimating groundwater balance at the end of May.
- Appreciate the participant and explain the process.
- The projected balance of groundwater at the end of May = Recharge (June-October rainfall + projected rainfall + secondary water bodies) – Draft (Kharif + projected water requirement).
- Encourage the participants to provide the data on the above items and estimate the projected groundwater balance at the end of May.

7. Short Study: Water quality analysis Time: 15 min.

Objective : Participants discuss the importance of water analysis.

Rationale : Majority of the villages do not know whether the water available from the local borewell is fit for drinking. It is important to know the quality of the water that they drink. This will help them to take necessary precautions. This exercise helps participants understand the importance of water quality analysis.

Materials : Two transparent glasses, 20 grams of alum, one litre drinking water of the village and one liter safe drinking water.

Process

- Fill one glass with drinking water of the village.
- Fill the second glass with safe drinking water.
- Grind the alum to make powder and add it equally to the two glasses.
- Observe the two glasses after 30 minutes.
- Impurities precipitate in the first glass which is filled with water from the village.
- Discuss the measures/precautions to be taken for ensuring access to safe drinking water.

8. Short Study: Cooperation among committee members Time: 15 min.

Objective : Participants will enact a role-play and discuss the need for cooperation among committee members in a GMC/HUN in decision making and implementation

Rationale : Collective decision-making is very important for effective management of groundwater resources. Decisions taken by the committee can be implemented successfully when there is a firm backing from its members and also by every groundwater user. The following role-play will help participants understand the need for cooperation among committee members.

Exercise : Use of water

Materials : Three transparent bottles with water

Role Play

- The first person introduces himself as president of the village committee and informs the villagers that due to mechanical failure, there would be no drinking water supply for the coming two days and there is a need to use water judiciously.
- Next, three persons carry three bottles of water to their work place and act as laborers.
- After a while, the first person drinks some water and washes his face with the remaining water.
- The second person does the same as the first person and empties the bottle.
- The third person drinks little and stores the rest of the water.
- After some time, the first person approaches the second person for water. The second person shows his empty bottle and expresses that he too is thirsty.
- Then the two persons go to the third person and drink from his bottle.
- After sometime, all the three feel thirsty and express their anguish showing the empty bottles in their hands.

Preparation

- Select three persons, explain the play and its importance to them and ask them to enact in the session.
- Compare the water bottles with groundwater and discuss the importance of unity and work together in managing the groundwater resources.

9. Group dynamic: communication distortion Time: 15 min.

Objective :

- Illustrate the breakdown of communication.
- Demonstrate the importance of good communication in undertaking community projects.

Materials : None

Process

- Ask all the participants to form a circle.
- The facilitator then whispers a message to the first person on his right or to his left. Pass on the message on, i.e., whisper to the next person only once and the next until the message gets to the other end of the circles.
- Request the participants not to repeat the message.
- Ask the last person to receive the message to say the sentence aloud. The first person to whom the facilitator whispered the message will verify the accurateness or correctness of the message.
- Relate the activity to good and clear communication as a significant factor in successfully carrying out community undertakings. People may view the degree of change in the original message or breakdown in communication as changes caused by certain hindrances or barriers to effective communication that affects implementation of community projects.

10. Evaluation Time: 10 min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- List the data required to estimate groundwater draft?
- How do you estimate the water requirement for a crop?
- What data is required to estimate the average projected rainfall during November to May in a given year?
- What are the factors to be considered during the estimation of recharge through secondary water bodies?